Experiment-5 Atwood's Machine

In this part, we will use Atwood's machine arranged on an air table to study Newton's second law of motion.

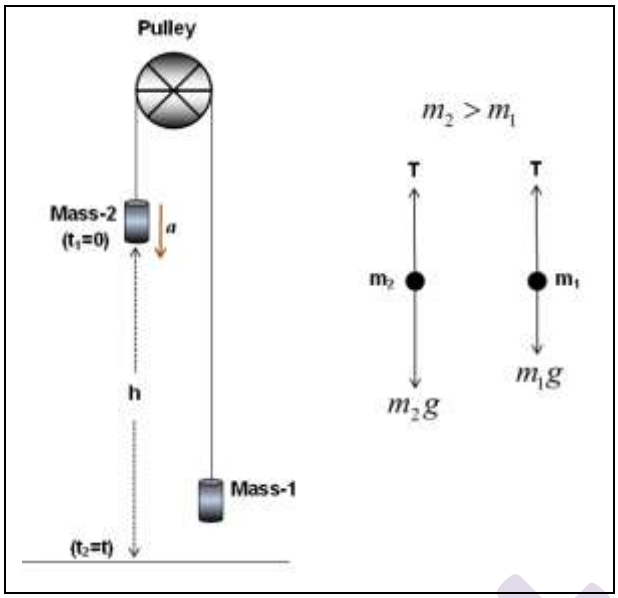


Figure-14: Atwood's machine and free-body diagram of the forces on the each mass.

An Atwood's machine is a physical device used to measure the acceleration of a mass system due to gravity, and to study the dependence of acceleration on the net force as well as on the total mass of a system. In order to move an object that is initially at **rest**, we must apply a force on it. The acceleration of an object depends on the net applied force, and the mass of the object.

The system shown in Figure-(14) is called an Atwood's machine. It consists of two different masses m_1 and m_2 ($m_2 > m_1$) connected at the ends of a string passing over a pulley. The forces acting on each of the masses are also shown in the figure. When the system of two masses is released from the **rest**, the heavier mass m_2 moves downward with **constant** acceleration and the mass m_1 moves upward with the **same acceleration**. Since m_2 accelerates downwards, it experiences a resultant force in this direction such that $m_2 g > T$. As for the mass m_1, we have $m_1 g < T$.

Units in Newton's Second Law

Force	Mass	Acceleration
Newton (N)	Kilogram (kg)	m/s^2

The unit of force is called the **Newton** (**N**). One newton is the amount of net force that gives an acceleration of 1 meter per second squared to a body with a mass of 1 kilogram. So it is related to the units of mass, length, and time (1**N**=1kg.m/s^2).

In the free body diagram of the Atwood's machine; **T** is the tension in the string (cord), m_1 is the **lighter mass**, m_2 is the **heavier mass**, and **g** is the acceleration due to **gravity**. Assuming that there is no friction on the system;

- The net force on m_1 is the difference between the **tension** (T) and $m_1 g$; $T > m_1 g$.
- The net force on the mass m_2 is the difference between the **tension** and $m_2 g$ such that $T < m_2 g$.

The acceleration of an object is directly proportional to the net force acting on it, and is inversely proportional to the object's mass. The direction of the acceleration is in the direction of the net force acting on the object. This is Newton's second law of motion. Newton's second law can be written as the equation:

$$\vec{a} = \frac{\Sigma \vec{F}}{m} \qquad (49)$$

where;

$\vec{a}$:	Acceleration of the object,
m :	Mass of the object,
$\Sigma \vec{F}$:	Net force on the object.

We can apply **Newton's second law** to the bodies of m_1 and m_2 on which the net force is **not** zero. So, the bodies are **not** in equilibrium and hence are accelerating. The net force on the each mass is equal to the mass of the body times its acceleration:

$$\Sigma \vec{F} = m \vec{a} \qquad (50)$$

Newton's second law relates the description of motion (acceleration) to the cause of motion (force). For m_2>m_1, Equation-(50) applied to each mass gives:

$$F_{Net} = T - m_1 g = m_1 a \quad (51)$$

$$F_{Net} = m_2 g - T = m_2 a \quad (52)$$

where;

T :	The tension in the string,
a :	The magnitude of the linear acceleration of **either mass** in the Atwood's machine,
g :	The acceleration due to **gravity**.

The magnitude of the force of the gravity on an object, mg, is called the object's weight.

In an Atwood's machine, the difference in **weight** between two hanging masses determines the **net force** ($\mathbf{F}_{Net}$) acting on the system of both masses. This net force accelerates both of the hanging masses; the **heavier mass** (m_2) is accelerated downward, and the **lighter mass** (m_1) is accelerated upward.

Combining Equations-(51) and (52) leads to:

$$T = m_1 a + m_1 g \quad (53)$$

$$T = m_2 g - m_2 a \quad (54)$$

$$m_1 a + m_1 g = m_2 g - m_2 a \quad (55)$$

$$m_1 a + m_2 a = m_2 g - m_1 g \quad (56)$$

$$\boxed{a = g\left(\frac{m_2 - m_1}{m_1 + m_2}\right)} \quad (57)$$

Equation-(57) gives the theoretical (expected) acceleration for the system of both masses. As seen from the expression, the expected acceleration derived from Newton's second law is equal to "g" times the difference in mass divided by the total mass. So, the net force on the mass system of the Atwood's machine is the difference between the two masses times the acceleration of gravity such that g(m_2-m_1).

If the mass, m_2 is released from rest (v_0=0) at a **vertical** distance h=y above the floor with the **initial** time t_1=0, m_2 strikes the floor at a **final** time t_2=t after its release. The relationship between the distance of the fall (y) and the linear acceleration (a) of the system:

$$y = y_0 + v_0 t + \frac{1}{2} a t^2 \quad (58)$$

Since the initial position of the mass, y_0=0 and the initial velocity, v_0=0, Equation-(58) becomes:

$$\boxed{y = \frac{1}{2} a t^2} \quad \textbf{(Experimental)} \quad (59)$$

The distance (y) through which each mass travels in a time interval (t), starting from rest, is given in terms of the acceleration (a) by the Equation-(59).

Obviously, since the acceleration (a) of the system is **constant** and since both pucks start motion from **rest**, the relation given by the Equation-(59) holds. Solving this expression for the acceleration (a) in terms of the measured quantities of the displacement (y) and the time interval (t) gives:

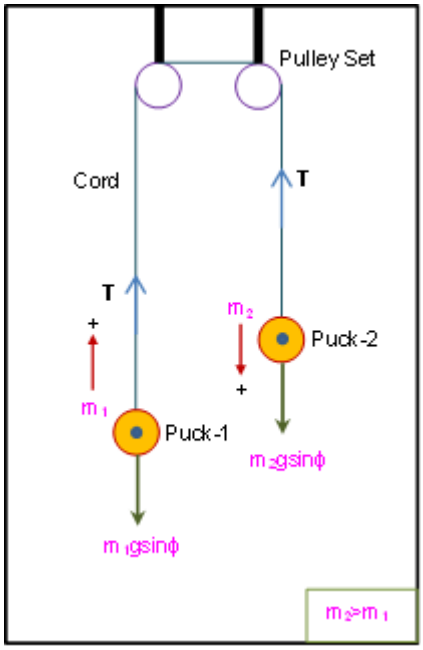
$$\boxed{a = \frac{2y}{t^2}} \quad \textbf{(Experimental)} \quad (60)$$

A graph of **y-t** for the motion with constant acceleration (a) is always a parabola that passes through the origin of that graph. However, if a graph of **y-t^2** is plotted, we will find a straight line which has a **slope** of:

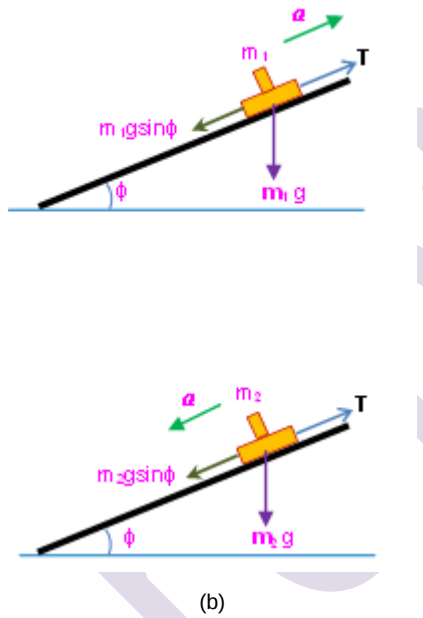
$$\boxed{slope = \frac{1}{2} a} \quad \textbf{(Experimental)} \quad (61)$$

and it will pass through the origin.

Equation-(61) will give the **acceleration** (a) of the Atwood's machine for the applied force of (m_2-m_1)g using a total mass (m_1+ m_2).



(a)



(b)

Figure-15: The experimental set-up of an **Atwood's machine** consisting of two pucks with the masses m_1 and m_2 on an inclined air table (a). The acceleration that is experienced by the both masses is a **constant** with the heavier mass (m_2) falling downwards and the lighter mass (m_1) rising (b).

To construct an Atwood's machine consisting of two pucks connected by a string on an **inclined** air table with **the angle of inclination** of ϕ, we set up the system shown in the Figure-(15a).

The two pucks act as the two masses required for the machine. Putting extra mass on it increases the mass of one of the puck. The greater mass (m_2) is shown in the Figure-(15b). Two forces along the **inclined plane** act on it:

- The tension **T** in the cord pulling upward and,
- The component of its **weight** $m_2 g \sin\phi$.

Since this mass accelerates downward direction, the tension **T** is smaller than $m_2 g \sin\phi$, and therefore the force acting on the mass m_2 is given by:

$$m_2 g \sin\phi - T = m_2 a \qquad (62)$$

Similarly, the net force acting on the mass m_1 is:

$$T - m_1 g \sin\phi = m_1 a \qquad (63)$$

By adding the two equations side by the side and the eliminating, we can find the acceleration:

$$a = \frac{(m_2 - m_1)}{(m_1 + m_2)} g \sin\phi \qquad \textbf{(Expected)} \qquad (64)$$

So, experimental measurement of the distance (***y***) and time (**t**) therefore determines the **acceleration** (***a***), whose value is the result of this experiment. The acceleration (***a***) is related to the acceleration (**g**) due to **gravity** by the Equation-(64).

Note that if the angle of the inclination of the air table is $\phi = 90^0$, then we get:

$$a = \frac{(m_2 - m_1)}{(m_1 + m_2)} g \qquad (65)$$

Solving for the tension in the cord, we can also obtain:

$$T = \frac{2 m_1 m_2}{(m_1 + m_2)} g \sin\phi \qquad \text{(Experimental)} \qquad (66)$$

where, g is the acceleration due to gravity (=9.8m/s^2) and ϕ is the angle that the air table makes with the horizontal (the angle of the inclination).

6.1. *Experimental Procedures*

This part of the experiment is carried out on the inclined air table.

1. First, level the air table off, then put a block under the single leg of the air table by raising its **backside** for an inclined position with the angle:

   - $\phi=20^0$

   You can use an angle finder to measure the inclination angle.

2. Measure the masses m_1 and m_2.

3. Mount the pulley set to the upper side of the inclined air table and place the string (cord) over the both pulleys (see Figure-14a).

   3.1. Arrange the length of the string so that either of the pucks m_1 or m_2 can move at least **20cm** (0,2 meters) on the inclined plane.
   3.2. Remove the air hoses of the pucks and then:
   - Add additional mass to the puck on the right and **record** the total mass of the puck with **extra mass** as m_2.
   - Attach the rings to the two pucks.

   3.3. Connect again the air hoses to the pucks.
   3.4. Then, attach the both pucks to the ends of the **string** passing over the pulleys by using the rings.

   - The distance between the centers of the two pulleys mounted on the upper side of the inclined air table should be **30cm**.
   - The string passing over the pulleys which is attached to the both pucks **should be** parallel to the plane of the air table.

4. Place the puck on the left (with smaller mass, m_1) at the lowest position and the other puck at the highest position.

   Note that the puck, m_2 on the right has extra mass (that is, greater mass).

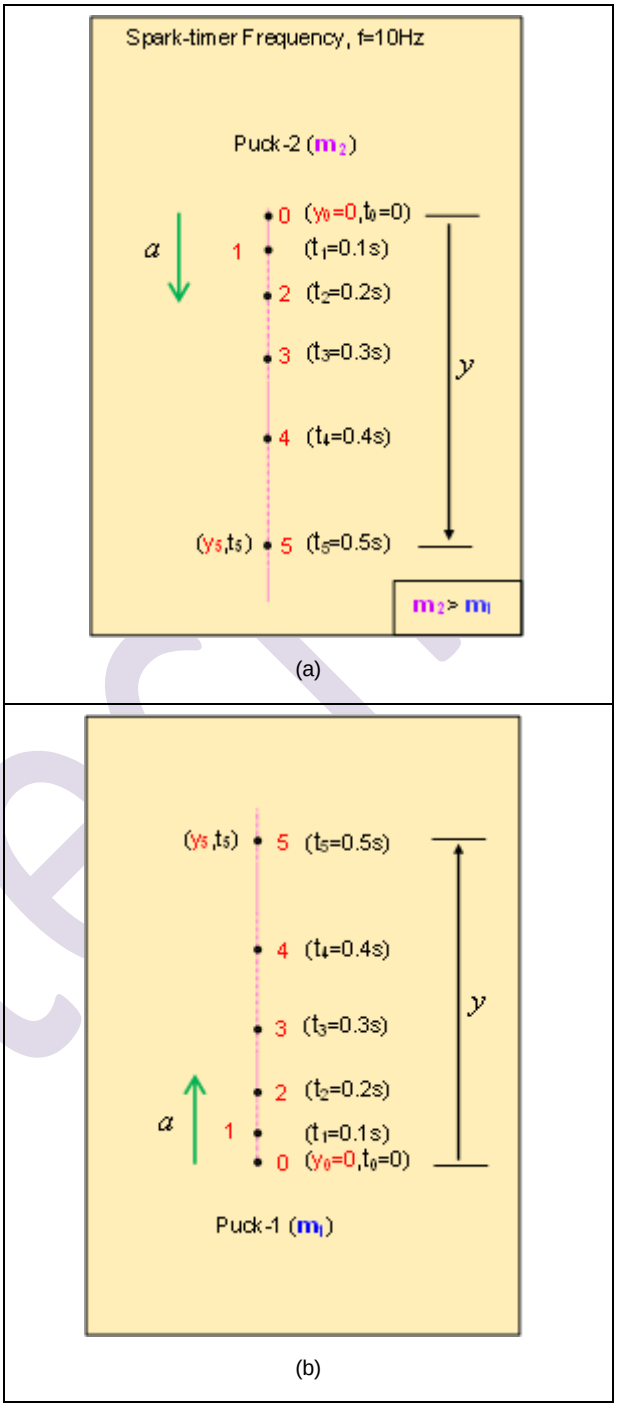


Figure-16: The dots produced by the Puck-**2** (**a**) and Puck-**1** (**b**) on the data sheet in an **Atwood's machine**.

5. Activating **only** the **P**ump, footswitch (P) for the pucks, **release** the system from rest and observe the motion of the two pucks. Repeat this several times to get familiar with the motion.

6. The dots produced on the data sheet will look like those shown in the **Figure-(16)**.

7. Now, set the spark-timer frequency to:

   - f=10Hz.

   which makes a mark *every 1/10 second*. You may, however, change this to 20Hz if your data was not convenient.

8. Repeat now by putting the **two** foot switches on top of each other and depressing them simultaneously to produce the spark record for the both pucks. Observe that the puck on the right goes down while the other goes up the incline.

9. **Turn off the spark-timer and then remove the pucks from the air table.**

10. Remove the data sheet and observe the dots produced on it.

    - ***What kind of a trajectory do the pucks have?.***
    - ***Can you say that both pucks have the same motion?. Explain your answer.***

11. **Starting from the first dot**, number the dots produced by each puck on the data sheet as **0,1,2,3,4, and 5**.

    11.1. The first dot of each trajectory on the data sheet is to be taken as **dot 0**.
    11.2. Use this point (dot **0**) as the reference point for zero position and zero time.
    11.3. Taking the **positive** y-axis to be the direction of motion, **measure** the position and the time of five data points in each trajectory relative to the **dot-0** (0.05 second between successive two dots since f=20Hz).
    11.4. Fill the data values for the position and time in data Table-(8).

Record the position and time measurements of **five** data points of the pocks m_1 and m_2 relative to the **dot 0** in the data Table.

12. Use the data to **plot** the **y** versus t^2 graph for **either** of the masses m_1 or m_2.

The experimental acceleration (a) can be determined by measuring the time (t) required for the mass (m_2) to fall from rest a specific distance (y) as seen in the Figure-16. The acceleration of the each mass is the **same** since they are connected by the string. You should obtain a linear relation between the position (y) and the square of time (t^2) for either of the masses m_1 or m_2. If not, collect the data again with the same settings.

12.1. Now, **draw** the best line passing through the data values on the graph. Be careful with the units of vertical and horizontal axis of the graph.
12.2. Find the **slope** (***s***) of the best-fit line (straight line) and record it in the data Table.
12.3. Find the acceleration (***a***) of the puck from the **slope** of the graph.

13. Calculate experimental value of the **acceleration** (**g**) **due to gravity** (with the Atwood's Machine), in units of **m/s^2**. Compare this with **g**=9.80 m/s^2 to obtain a percentage difference. For this, use:

    - The acceleration (***a***) you obtained,
    - Sinϕ and,
    - The masses m_1 and m_2 in the Equation-(64).

    Show the details of the calculations.

14. Determine the **tension** (**T**) in the cord with the correct unit by using Equation-(66).

15. Find also the **weight** of the mass (m_2). Note that the weight is a force, the pull of gravity acting on the mass of puck (m_2).

16. Show that the **tension** in the cord (string) is smaller than the force (**weight**) along the **inclined plane** since the mass (m_2) accelerates downward.

17. Is your value of the **acceleration** (**g**) consistent with the accepted value of **g**=9.80m/s^2. If your comparison shows a percent difference more than 10%, explain at least one of the possible error sources.

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6.2. LABORATORY REPORT

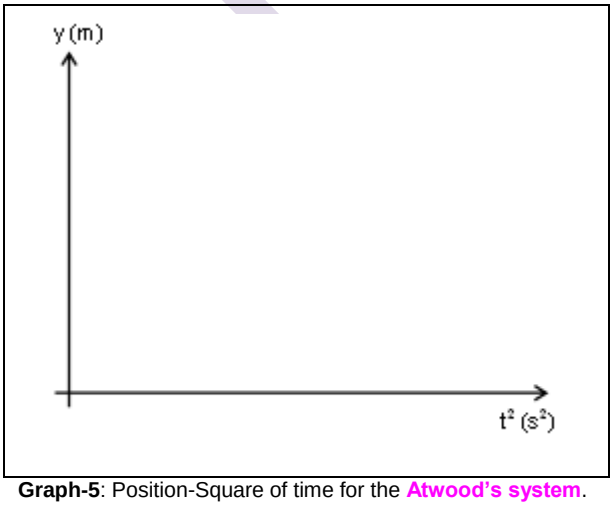
Atwood's Machine

Table-8: Position and time measurements of five data points of the pucks $\mathbf{m_1}$ and $\mathbf{m_2}$ for the **Atwood's system**.

Dot Number	Measured				Experimental
	Position		Time	Square of Time	Slope
	Mass,$\mathbf{m_2}$	Mass,$\mathbf{m_1}$			
	y(m)	y(m)	t(sec)	$t^2(s^2)$	$s(m/s^2)$
0	0	0	0	0	.....
1	.....	.....	.....	.....	
2	.....	.....	.....	.....	
3	.....	.....	.....	.....	
4	.....	.....	.....	.....	
5	.....	.....	.....	.....	

Record the position and time measurements of five data points of the pucks m_1 and m_2 relative to the dot 0.

Angle of inclination, ϕ(deg):	**20**
Spark-timer frequency, f(Hz):	**10**
Puck,m_1(kg):	.....
Puck with extra mass,m_2(kg):	.....
Slope, s(m/s^2):	.....
Acceleration, a(m/s^2):	.....
Acceleration due to gravity, g(m/s^2):	.....
Tension in the cord, T(N):	.....
Weight of the puck with extra mass, W(N):	.....



Graph-5: Position-Square of time for the **Atwood's system**.

Acceleration due to gravity (Experimental)	Acceleration due to gravity (Expected)	Percent Error (Calculated)
g(m/s^2)	g'(m/s^2)	Δg (±%)
.....	.....	.....

Conclusions and Questions

- Describe briefly the basic concept of the Atwood's machine. What is the **net force** which causes an acceleration of the system?.
- How can you determine the **experimental** value of the constant acceleration from the distance and time measurements?. Show your work.
- How can you calculate the **expected** (theoretical) value of the acceleration from the masses and mass differences?.
- What is the relationship between the **acceleration** and the **total mass** when the net force is held constant?.
- If the **net force** due to mass difference remains constant but the **total mass** increases, what happens to the acceleration?.
- How does the acceleration of the **greater mass** (right mass) compare to that of the left mass?.
- Is the acceleration **same** for both masses?. Write a detailed **conclusion** about what you have learned in this experiment.

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